

Algebraic Geometry and Coding Theory

Lecture 4

October 8

Main theme. The zeta function packages the arithmetic of a curve into one generating function. Riemann–Roch makes it rational and gives its functional equation. The Hasse–Weil bounds from Lecture 3, applied over every finite extension, then imply Weil’s Riemann hypothesis. Finally, a positive averaging argument gives the sharper asymptotic bound $A(q) \leq \sqrt{q} - 1$.

Throughout, $X/\mathbf{F}_q$ is a smooth projective geometrically integral curve of genus g , and $F = \mathbf{F}_q(X)$ has exact constant field $\mathbf{F}_q$. We distinguish effective-divisor counts A_n , closed-point counts B_r , and extension-field point counts $N_m = \#X(\mathbf{F}_{q^m})$. The polynomial $L_F(t)$ is not the vector space $L(D)$.

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1 From the classical zeta function to curves

The opening of Morandi's Chapter 7 passes from integers to ideals and then to divisors. The common operation is to sum over objects that have a unique decomposition into prime pieces. This gives both a series and an Euler product. We review that motivation before developing the curve case in detail.

1.1 Integers and the Euler product

For a complex number s with $\operatorname{Re}(s) > 1$, the classical Riemann zeta function is

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \quad n^{-s} = \exp(-s \log n). \quad (1)$$

If $\sigma = \operatorname{Re}(s) > 1$, then $|n^{-s}| = n^{-\sigma}$, so the series is absolutely convergent. This follows from the integral test for $\sum n^{-\sigma}$.

Proposition 1.1 (Euler product). *For $\operatorname{Re}(s) > 1$,*

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}. \quad (2)$$

Proof. For a finite set S of primes, expand each factor as a geometric series:

$$\prod_{p \in S} (1 - p^{-s})^{-1} = \prod_{p \in S} \left(\sum_{e=0}^{\infty} p^{-es} \right) = \sum_{\substack{n \geq 1 \\ \text{all prime factors of } n \text{ lie in } S}} n^{-s}.$$

Unique prime factorization says that each integer in the last sum occurs exactly once. Let S consist of all primes at most B . An integer omitted from that sum has a prime factor greater than B , and hence is itself greater than B . The absolute value of the omitted part is therefore at most

$$\sum_{n > B} n^{-\sigma},$$

which tends to zero. Passing to the limit proves the identity. $\square$

The factor indexed by p permits every multiplicity $0, 1, 2, \dots$ of that prime. The constant term 1 corresponds to the integer 1, whose prime factorization is empty. This normalization will reappear as the contribution of the zero divisor.

The classical Riemann hypothesis concerns the meromorphic continuation of (1): its assertion is that every nontrivial zero has real part $1/2$. Here “nontrivial” excludes the zeros at the negative even integers. The analytic continuation and the facts about these classical zeros are background motivation, not inputs to our arguments for curves. In the curve setting, rationality will give the continuation directly.

Remark 1.2 (A sign convention in the reference). The Euler factors in (2) involve p^{-s} . The corresponding displays in [M, §7.1] omit this minus sign. We use the convention compatible with the defining series $\sum n^{-s}$.

1.2 Number fields and the ideal–divisor analogy

Every nonzero ideal of $\mathbf{Z}$ has the form (n) for a unique positive integer n , and

$$|\mathbf{Z}/(n)| = n.$$

Thus the classical zeta function can also be written as a sum over nonzero ideals, weighted by the sizes of their quotient rings.

For a number field K , let $\mathcal{O}_K$ be its ring of integers. For a nonzero integral ideal $I \subseteq \mathcal{O}_K$, its norm is

$$N(I) = |\mathcal{O}_K/I|.$$

The analogous series is

$$\zeta_K(s) = \sum_{0 \neq I \subseteq \mathcal{O}_K} N(I)^{-s}. \quad (3)$$

The ideal $I = \mathcal{O}_K$ is included and contributes 1.

The relevant algebraic fact is that $\mathcal{O}_K$ is a Dedekind domain: its nonzero ideals have unique factorizations

$$I = \prod_{\mathfrak{p}} \mathfrak{p}^{e_{\mathfrak{p}}}, \quad e_{\mathfrak{p}} \geq 0,$$

with finite support, and the norm is multiplicative. Consequently, the counterpart of the Euler product is

$$\zeta_K(s) = \prod_{\mathfrak{p}} (1 - N(\mathfrak{p})^{-s})^{-1} \quad (\operatorname{Re}(s) > 1).$$

We use these number-field facts only for motivation, as in [M, §7.2]; no theorem about number-field zeta functions is needed below.

Allowing arbitrary integer exponents gives the group of fractional ideals. A fractional ideal is a nonzero $\mathcal{O}_K$ -submodule of K that becomes an integral ideal after multiplication by some nonzero element of $\mathcal{O}_K$. Quotienting this group by the principal fractional ideals gives the ideal class group.

On a curve, the corresponding additive objects are

$$D = \sum_P n_P P, \quad n_P \in \mathbf{Z},$$

where P ranges over closed points. Effective divisors have $n_P \geq 0$; principal divisors are the divisors (f) of nonzero rational functions. Accordingly, the dictionary is

nonzero prime ideals	$\longleftrightarrow$	closed points
nonzero integral ideals	$\longleftrightarrow$	effective divisors
fractional ideals	$\longleftrightarrow$	arbitrary divisors
ideal classes	$\longleftrightarrow$	divisor classes.

The multiplication of ideals becomes addition of divisors. This is the motivation for the next definition.

2 Divisors, point counts, and the zeta function

2.1 Effective divisors and the Euler product

Recall that a closed point P of degree r is a Frobenius orbit of r geometric points, or equivalently a point with residue field $\kappa(P) \simeq \mathbf{F}_{q^r}$. Define

$$N(P) = |\kappa(P)| = q^{\deg P}.$$

For an effective divisor $E = \sum_P e_P P$, put

$$N(E) = \prod_P N(P)^{e_P} = q^{\deg E}.$$

In particular, $N(0) = 1$.

Definition 2.1. Let

$$A_n = \#\{E \geq 0 : \deg E = n\} \quad (n \geq 0).$$

The zeta generating function and the zeta function in the variable s are

$$Z_F(t) = \sum_{n=0}^{\infty} A_n t^n = \sum_{E \geq 0} t^{\deg E}, \quad \zeta_F(s) = Z_F(q^{-s}). \quad (4)$$

We first regard $Z_F(t)$ as a formal power series. Its convergence for $|t| < q^{-1}$, and hence the convergence of $\zeta_F(s)$ for $\operatorname{Re}(s) > 1$, will be proved in Section 3.

Lemma 2.2. *There are finitely many effective divisors of each fixed degree. Moreover,*

$$A_0 = 1, \quad A_1 = \#X(\mathbf{F}_q) = N_1.$$

Proof. For each r , there are only finitely many points in $X(\mathbf{F}_{q^r})$, and thus finitely many closed points of degree r . An effective divisor of degree n involves only points of degrees at most n , and each coefficient is at most n . There are consequently only finitely many possibilities.

The only effective divisor of degree zero is 0. An effective divisor of degree one must be one degree-one closed point with coefficient one, namely an $\mathbf{F}_q$ -rational point. $\square$

Remark 2.3 (The zero divisor is included). The sums in (4) begin at $n = 0$. Omitting the zero divisor replaces $Z_F(t)$ by $Z_F(t) - 1$ and destroys the Euler-product normalization. This corrects the starting index in the initial displays of [M, §7.3, p. 67], consistently with his Corollary 7.6 and Theorem 7.18. It is also the convention used in Lecture 3.

Proposition 2.4 (Euler product for a curve). *As a formal power series,*

$$Z_F(t) = \prod_{P \text{ closed}} (1 - t^{\deg P})^{-1}. \quad (5)$$

Proof. Expand the factor for P as

$$1 + t^{\deg P} + t^{2 \deg P} + \dots.$$

Selecting the term $t^{e_P \deg P}$ from each factor amounts to selecting an effective divisor $\sum_P e_P P$. The resulting monomial is $t^{\deg E}$. Every effective divisor has a unique such expression, so its contribution occurs exactly once. For each coefficient, only finitely many factors matter, which justifies the identity formally. $\square$

2.2 Extension-field points and the exponential expression

The coefficient A_n counts divisors, not points over $\mathbf{F}_{q^n}$. To relate the two counts, introduce

$$B_r = \#\{P \text{ closed} : \deg P = r\}, \quad N_m = \#X(\mathbf{F}_{q^m}).$$

Lemma 2.5 (Frobenius-orbit counting). *For every $m \geq 1$,*

$$N_m = \sum_{r|m} r B_r. \quad (6)$$

In particular, $N_m \geq N_1$ for all $m \geq 1$.

Proof. A Frobenius orbit of length r consists of points fixed by the m -th power of Frobenius exactly when $r \mid m$. In that case all its r geometric points are $\mathbf{F}_{q^m}$ -rational. Otherwise it contributes no points. Summing over the orbits gives the formula.

The term $r = 1$ contributes $B_1 = N_1$, and every other term is nonnegative. Equivalently, every $\mathbf{F}_q$ -rational point is still rational after any constant-field extension. $\square$

For example,

$$N_2 = B_1 + 2B_2, \quad A_2 = \binom{B_1 + 1}{2} + B_2.$$

The first formula counts geometric points fixed by Frobenius squared. In the second, a degree-two divisor is either an unordered pair of rational points, with repetition allowed, or one closed point of degree two.

Proposition 2.6 (Exponential expression). *We have the formal power-series identity*

$$Z_F(t) = \exp \left(\sum_{m=1}^{\infty} N_m \frac{t^m}{m} \right). \quad (7)$$

Proof. Since $Z_F(0) = 1$, its formal logarithm is defined. Group the Euler factors by degree and use $-\log(1 - u) = \sum_{k \geq 1} u^k / k$:

$$\begin{aligned} \log Z_F(t) &= - \sum_{r \geq 1} B_r \log(1 - t^r) \\ &= \sum_{r \geq 1} \sum_{k \geq 1} B_r \frac{t^{rk}}{k} \\ &= \sum_{m \geq 1} \left(\sum_{r|m} r B_r \right) \frac{t^m}{m} \\ &= \sum_{m \geq 1} N_m \frac{t^m}{m}. \end{aligned}$$

In the coefficient of t^m , only divisors r of m appear, so the rearrangement is finite coefficient by coefficient. Exponentiating proves the proposition. $\square$

Thus Z_F contains two kinds of counting data: its coefficients are effective-divisor counts, while the coefficients of its logarithm are normalized extension-field point counts. This is the bridge to the power sums that will appear later.

3 Riemann–Roch and rationality

We now develop the class-counting argument of [M, §7.3]. Its input is Riemann–Roch, not a bound on the zeros of Z_F . Recall

$$L(D) = \{f \in F^\times : (f) + D \geq 0\} \cup \{0\}, \quad \ell(D) = \dim_{\mathbf{F}_q} L(D),$$

and, for a canonical divisor K_X ,

$$\ell(D) - \ell(K_X - D) = \deg D + 1 - g, \quad \deg K_X = 2g - 2. \quad (8)$$

In particular, $\ell(D) = \deg D + 1 - g$ whenever $\deg D > 2g - 2$.

3.1 The finite degree-zero class group

Two divisors are linearly equivalent, written $D \sim E$, if $D - E$ is principal. Let

$$\mathrm{Cl}(F) = \mathrm{Div}(X) / \mathrm{Prin}(X), \quad \mathrm{Cl}^0(F) = \{[D] \in \mathrm{Cl}(F) : \deg D = 0\}.$$

Degree is well defined on divisor classes because principal divisors have degree zero. For any integer n , write

$$\mathrm{Cl}^n(F) = \{[D] : \deg D = n\}.$$

This is a set, rather than a subgroup unless $n = 0$.

Lemma 3.1 (Finiteness of the class number). *The group $\mathrm{Cl}^0(F)$ is finite. Write*

$$h = |\mathrm{Cl}^0(F)|.$$

Whenever $\mathrm{Cl}^n(F)$ is nonempty, it has exactly h elements.

Proof. Choose an effective divisor E of degree $e \geq g$; a sufficiently large multiple of any closed point will do. Translation by $[E]$ gives a bijection

$$\mathrm{Cl}^0(F) \longrightarrow \mathrm{Cl}^e(F), \quad [D] \longmapsto [D + E].$$

For every divisor C of degree e , Riemann–Roch gives

$$\ell(C) \geq e + 1 - g \geq 1.$$

Choose a nonzero $f \in L(C)$. Then $C + (f)$ is an effective divisor of degree e representing $[C]$. There are only finitely many effective divisors of degree e , so there are finitely many classes in $\mathrm{Cl}^e(F)$, and hence in $\mathrm{Cl}^0(F)$.

For any n admitting a divisor D_n , translation by $[D_n]$ similarly identifies $\mathrm{Cl}^0(F)$ with $\mathrm{Cl}^n(F)$. $\square$

There is one issue to retain for now: not every integer has yet been shown to occur as a divisor degree. The image of the degree homomorphism is

$$\deg \mathrm{Div}(X) = \partial \mathbf{Z}, \quad \partial = \gcd\{\deg P : P \text{ closed}\} \geq 1. \quad (9)$$

Thus $\mathrm{Cl}^n(F)$ is nonempty precisely when $\partial \mid n$. In particular, $A_n = 0$ if $\partial \nmid n$. We will prove $\partial = 1$ in Section 4; this must not be replaced by an unproved assumption that X has a rational point.

3.2 Counting the effective divisors in one class

Lemma 3.2 (Morandi's class-counting formula). *For any divisor D ,*

$$\#\{E \geq 0 : E \sim D\} = \frac{q^{\ell(D)} - 1}{q - 1}. \quad (10)$$

Consequently, if $n \geq 0$, $n > 2g - 2$, and $\partial \mid n$, then

$$A_n = \frac{h}{q - 1} (q^{n+1-g} - 1). \quad (11)$$

Proof. The map

$$L(D) \setminus \{0\} \longrightarrow \{E \geq 0 : E \sim D\}, \quad f \longmapsto D + (f),$$

is surjective by the definition of linear equivalence. Two functions f and f' give the same divisor exactly when $(f/f') = 0$. A rational function with no zeros or poles on a projective geometrically integral curve is a nonzero constant, so this happens exactly when $f/f' \in \mathbf{F}_q^\times$.

Thus each fiber has $q - 1$ elements. There are $q^{\ell(D)} - 1$ nonzero functions in $L(D)$, proving (10). Equivalently, the effective divisors in the class are parametrized by the one-dimensional subspaces of $L(D)$.

In degree $n > 2g - 2$, Riemann–Roch makes $\ell(D) = n + 1 - g$ independent of the class. There are h classes of this degree when $\partial \mid n$. Adding their equal contributions proves (11). $\square$

This is the essential use of Riemann–Roch in rationality. After a finite number of initial degrees, the coefficient A_n is a fixed linear combination of q^n and 1 on the allowed degrees. Its tail can therefore be summed by geometric series.

3.3 Summing the tail and locating a simple pole

Proposition 3.3 (Rationality before proving $\partial = 1$). *The series $Z_F(t)$ converges for $|t| < q^{-1}$ and is the expansion of a rational function. This rational function has a simple pole at $t = 1$ and at $t = q^{-1}$.*

Proof. Choose a nonnegative multiple b of ∂ with $b > 2g - 2$, and let

$$P_{<b}(t) = \sum_{0 \leq n < b} A_n t^n.$$

The allowed degrees at least b are $b, b + \partial, b + 2\partial, \dots$. Formula (11) gives

$$\begin{aligned} Z_F(t) &= P_{<b}(t) + \frac{h}{q - 1} \sum_{j \geq 0} (q^{b+j\partial+1-g} - 1) t^{b+j\partial} \\ &= P_{<b}(t) + \frac{h}{q - 1} \left(\frac{q^{b+1-g} t^b}{1 - (qt)^\partial} - \frac{t^b}{1 - t^\partial} \right). \end{aligned} \quad (12)$$

Both geometric series converge when $|t| < q^{-1}$. This proves convergence and rationality without making any assumption about ∂ .

At $t = 1$, the first fraction is regular, whereas the second has a simple pole: its numerator is nonzero, and $1 - t^\partial$ has a simple zero. The polynomial term cannot cancel that pole. At $t = q^{-1}$, the roles of the two fractions are reversed, with the same conclusion. $\square$

The same proof applies to any constant-field extension, because the extended curve is again smooth, projective, and geometrically integral. We will use its simple pole at 1 shortly.

For completeness, the Euler product is also an analytic identity on $|t| < q^{-1}$ and is nonzero there. Indeed, every closed point is itself an effective divisor, so for $0 \leq u < q^{-1}$,

$$\sum_P u^{\deg P} \leq \sum_{n \geq 1} A_n u^n < \infty.$$

For $|t| = u < q^{-1} < 1$, we then have

$$\sum_P \sum_{k \geq 1} \frac{|t|^{k \deg P}}{k} \leq \frac{1}{1-u} \sum_P u^{\deg P} < \infty.$$

Hence the logarithms of the Euler factors converge absolutely, and their exponential is a nonvanishing analytic function. Its power series is $Z_F(t)$ by (5).

4 Constant-field extensions and degree-one divisors

We follow the base-extension and pole-order argument in [M, Lemma 7.11, Proposition 7.12, and Corollary 7.13]. This establishes $\partial = 1$ without invoking the Hasse–Weil bound or Weil’s RH.

4.1 Splitting closed points and a product identity

For $r \geq 1$, let

$$X_r = X \times_{\mathbf{F}_q} \mathbf{F}_{q^r}, \quad F_r = \mathbf{F}_{q^r}(X_r),$$

and denote its zeta generating function by $Z_{F_r}(u)$. Degrees in Z_{F_r} are measured over $\mathbf{F}_{q^r}$, not over $\mathbf{F}_q$.

Lemma 4.1 (Splitting under a constant-field extension). *A closed point of degree m on X splits into $d = \gcd(m, r)$ closed points on X_r , each of degree m/d over $\mathbf{F}_{q^r}$.*

Proof. Write the original orbit as a cycle of length m under the q -power Frobenius σ . The Frobenius for the enlarged field is σ^r . On this cycle it moves forward by r positions. There are $\gcd(m, r) = d$ cycles for this action, each with m/d elements. These are exactly the closed points over $\mathbf{F}_{q^r}$ lying above the original point. $\square$

For instance, a degree-six point becomes two degree-three points after extending to $\mathbf{F}_{q^2}$, and six rational points after extending to $\mathbf{F}_{q^6}$. The degrees in these statements are relative to the new fields.

Proposition 4.2 (Constant-field product formula). *Let $\mu_r = \{\xi \in \mathbf{C} : \xi^r = 1\}$. Then*

$$Z_{F_r}(t^r) = \prod_{\xi \in \mu_r} Z_F(\xi t) \tag{13}$$

as formal power series and as rational functions.

Proof. Fix a closed point P of degree m , and put $d = \gcd(m, r)$. Its contribution to the Euler product on the left is

$$(1 - t^{rm/d})^{-d},$$

by the splitting lemma. Its contribution on the right is

$$\prod_{\xi \in \mu_r} (1 - \xi^m t^m)^{-1}.$$

As ξ runs through μ_r , the numbers ξ^m run through all (r/d) -th roots of unity, each d times. The elementary factorization

$$\prod_{\eta^{r/d}=1} (1 - \eta x) = 1 - x^{r/d}$$

therefore identifies these contributions. Multiplying over all closed points proves the formal identity. Rationality then makes it an identity of rational functions as well. $\square$

4.2 The degree map is surjective

Theorem 4.3 (Existence of a divisor of degree one). *For a smooth projective geometrically integral curve over $\mathbf{F}_q$,*

$$\partial = 1.$$

Equivalently, the degree map $\text{Div}(X) \rightarrow \mathbf{Z}$ is surjective.

Proof. Take $r = \partial$. Every effective divisor has degree divisible by r . Hence

$$Z_F(\xi t) = Z_F(t) \quad (\xi^r = 1).$$

The product formula (13) becomes

$$Z_{F_r}(t^r) = Z_F(t)^r.$$

The rational function $Z_{F_r}(u)$ has a simple pole at $u = 1$ by Section 3. Since $t \mapsto t^r$ has nonzero derivative r at $t = 1$ over $\mathbf{C}$, the left side has a simple pole there. The right side has a pole of order r , since $Z_F(t)$ itself has a simple pole at 1. Equality of pole orders gives $r = 1$. $\square$

Remark 4.4. A divisor of degree one need not be effective: it can have negative coefficients. Thus this theorem is not a proof that every curve has an $\mathbf{F}_q$ -rational point. It does show that every degree has exactly h divisor classes. The distinction between a degree-one divisor and a degree-one *effective* divisor is important in the rationality argument.

Corollary 4.5 (The genus-zero case). *If $g = 0$, then $h = 1$ and*

$$Z_F(t) = \frac{1}{(1-t)(1-qt)}. \quad (14)$$

Moreover, $X \simeq \mathbf{P}_{\mathbf{F}_q}^1$.

Proof. For a degree-zero divisor D , Riemann–Roch gives $\ell(D) \geq 1$. Thus $D + (f)$ is effective of degree zero for some $f \neq 0$, and must be zero. Every degree-zero divisor is principal, so $h = 1$.

Since $\partial = 1$ and $n > 2g - 2 = -2$ for all $n \geq 0$, formula (11) gives

$$A_n = \frac{q^{n+1} - 1}{q - 1}.$$

Summing yields

$$Z_F(t) = \frac{1}{q-1} \left(\frac{q}{1-qt} - \frac{1}{1-t} \right) = \frac{1}{(1-t)(1-qt)}.$$

To identify the curve, choose a divisor E of degree one. Riemann–Roch gives $\ell(E) = 2$, so some divisor equivalent to E is effective of degree one, hence is a rational point P . Now $\ell(P) = 2$, and a nonconstant function in $L(P)$ has just one simple pole. The degree of the associated map $X \rightarrow \mathbf{P}^1$ equals the degree of its pole divisor, so this map has degree one. A degree-one map between smooth projective curves is an isomorphism. This is the genus-zero argument recalled in [M, Corollary 7.14]. $\square$

5 The functional equation and the L -polynomial

We now have the class-counting formula in every sufficiently large degree. The remaining symmetry comes from replacing a divisor D by $K_X - D$ in Riemann–Roch.

5.1 Separating the finite part from the tail

Assume first that $g \geq 1$. Define

$$U(t) = \frac{1}{q-1} \sum_{\substack{[D] \in \text{Cl}(F) \\ 0 \leq \deg D \leq 2g-2}} q^{\ell(D)} t^{\deg D}, \quad (15)$$

$$V(t) = \frac{h}{q-1} \left(\frac{q^g t^{2g-1}}{1-qt} - \frac{1}{1-t} \right). \quad (16)$$

There are finitely many classes in the sum, so $U(t)$ is a polynomial of degree at most $2g-2$.

Lemma 5.1. *We have $Z_F(t) = U(t) + V(t)$.*

Proof. Sum (10) over all divisor classes of nonnegative degree. The contributions -1 give

$$-\frac{1}{q-1} \sum_{\deg[D] \geq 0} t^{\deg D} = -\frac{h}{q-1} \sum_{n \geq 0} t^n = -\frac{h}{(q-1)(1-t)},$$

because there are h classes of every degree. The contributions $q^{\ell(D)}$ from degrees at most $2g-2$ give $U(t)$. In the higher degrees, they give

$$\frac{h}{q-1} \sum_{n \geq 2g-1} q^{n+1-g} t^n = \frac{h}{q-1} \frac{q^g t^{2g-1}}{1-qt}.$$

These are precisely the two terms in $V(t)$. $\square$

Definition 5.2. The L -polynomial of $F/\mathbf{F}_q$ is

$$L_F(t) = (1-t)(1-qt)Z_F(t). \quad (17)$$

The formulas above show that $L_F(t)$ is a polynomial of degree at most $2g$. In genus zero it is 1. In every genus it has integer coefficients: multiplication of the series $\sum A_n t^n$ by $1 - (q+1)t + qt^2$ gives integer coefficients, and the higher ones vanish. Its constant coefficient is $L_F(0) = A_0 = 1$.

5.2 Riemann–Roch gives the functional equation

Theorem 5.3 (Functional equation). *The rational function Z_F and its numerator satisfy*

$$Z_F(t) = q^{g-1} t^{2g-2} Z_F\left(\frac{1}{qt}\right), \quad (18)$$

$$L_F(t) = q^g t^{2g} L_F\left(\frac{1}{qt}\right). \quad (19)$$

Proof. For $g = 0$, substitute (14) directly. Now suppose $g \geq 1$ and use $Z_F = U + V$.

The map $[D] \mapsto [K_X - D]$ permutes the classes whose degrees lie between 0 and $2g - 2$. If $n = \deg D$ and $E = K_X - D$, then $\deg E = 2g - 2 - n$, and Riemann–Roch gives

$$\ell(D) = n + 1 - g + \ell(E).$$

Consequently,

$$\begin{aligned} q^{\ell(D)} t^n &= q^{n+1-g+\ell(E)} t^n \\ &= q^{g-1} t^{2g-2} q^{\ell(E)} \left(\frac{1}{qt}\right)^{\deg E}. \end{aligned}$$

Summing over the finite set of classes gives

$$U(t) = q^{g-1} t^{2g-2} U\left(\frac{1}{qt}\right).$$

For the explicit rational tail, substitution gives

$$\begin{aligned} q^{g-1} t^{2g-2} V\left(\frac{1}{qt}\right) &= \frac{h}{q-1} \left(\frac{1}{t-1} - \frac{q^g t^{2g-1}}{qt-1} \right) \\ &= \frac{h}{q-1} \left(\frac{q^g t^{2g-1}}{1-qt} - \frac{1}{1-t} \right) = V(t). \end{aligned}$$

Adding proves (18).

Finally,

$$\left(1 - \frac{1}{qt}\right) \left(1 - \frac{1}{t}\right) = \frac{(1-t)(1-qt)}{qt^2}.$$

Multiplying (18) by $(1-t)(1-qt)$ and applying this identity proves (19). $\square$

This is the functional-equation argument in [M, Proposition 7.16 and Theorem 7.18]. It also gives an equation in the original complex variable:

$$\zeta_F(s) = q^{(g-1)(1-2s)} \zeta_F(1-s).$$

A symmetry of zeros is not yet a statement about their absolute values. That additional information will come from the Hasse–Weil estimates.

5.3 Coefficients, reciprocal roots, and all point counts

Theorem 5.4 (Structure of the L -polynomial). *The polynomial $L_F(t) \in \mathbf{Z}[t]$ has degree $2g$, constant coefficient 1, and leading coefficient q^g . Moreover,*

$$L_F(1) = h, \quad L_F(q^{-1}) = q^{-g}h, \quad (20)$$

and its coefficient of t is $N_1 - (q + 1)$. In particular, neither of the two denominator factors in

$$Z_F(t) = \frac{L_F(t)}{(1-t)(1-qt)} \quad (21)$$

is canceled.

Proof. We already know integrality, the degree bound, and the constant coefficient. Write

$$L_F(t) = \sum_{j=0}^{2g} a_j t^j,$$

allowing zero coefficients at the top until the degree is established. The functional equation gives

$$\sum_{j=0}^{2g} a_j t^j = \sum_{j=0}^{2g} q^{g-j} a_j t^{2g-j}.$$

Thus $a_{2g} = q^g a_0 = q^g \neq 0$, proving the degree assertion. More generally,

$$a_{2g-j} = q^{g-j} a_j \quad (0 \leq j \leq 2g).$$

For $g \geq 1$, multiply $U(t) + V(t)$ by the denominator to obtain

$$L_F(t) = (1-t)(1-qt)U(t) + \frac{h}{q-1} \left(q^g t^{2g-1}(1-t) - (1-qt) \right).$$

At $t = 1$, this equals h . Substituting $t = 1$ in the functional equation for L_F gives $L_F(q^{-1}) = q^{-g}h$. In genus zero, both assertions follow from $L_F = 1$ and $h = 1$.

Finally,

$$L_F(t) = (1 - (q+1)t + qt^2)(1 + N_1 t + \cdots),$$

so the coefficient of t is $N_1 - (q+1)$. For $g = 0$ that coefficient is zero, giving $N_1 = q+1$. Since h is a positive integer, the two values in (20) are nonzero. $\square$

Over $\mathbf{C}$, factor the numerator as

$$L_F(t) = \prod_{j=1}^{2g} (1 - \alpha_j t). \quad (22)$$

The numbers α_j are the *reciprocal roots*. The actual roots of $L_F(t)$, and hence the zeros of $Z_F(t)$, are α_j^{-1} . All roots are counted with multiplicity. The genus-zero product is empty and equals 1.

Proposition 5.5 (Point counts as power sums). *For every $m \geq 1$,*

$$N_m = q^m + 1 - \sum_{j=1}^{2g} \alpha_j^m. \quad (23)$$

Proof. Take formal logarithms of (21) using (22):

$$\begin{aligned} \log Z_F(t) &= \sum_{j=1}^{2g} \log(1 - \alpha_j t) - \log(1 - t) - \log(1 - qt) \\ &= \sum_{m \geq 1} \left(q^m + 1 - \sum_{j=1}^{2g} \alpha_j^m \right) \frac{t^m}{m}. \end{aligned}$$

Compare coefficients with the exponential expression (7). No root-size estimate has been used. $\square$

For $m = 1$, this recovers [M, Corollary 7.19]. The formula for every m is what we need for both RH and the asymptotic point bound. It also shows that constant-field extension to $\mathbf{F}_{q^r}$ replaces each reciprocal root α_j by α_j^r : the point counts over subsequent extensions are N_{rm} , so the corresponding numerator is $\prod_j (1 - \alpha_j^r u)$.

6 Hasse–Weil bounds imply Weil’s Riemann hypothesis

Morandi states the root-location theorem and then deduces the Hasse–Weil bound. Here we reverse that implication, using the independent Hodge-index argument recalled from Lecture 3. The power-sum argument below follows the supplementary note *Hasse_Weil_Bound_to_RH*.

6.1 Recall: Hodge index bounds every extension-field count

For every $m \geq 1$, the surface argument from Lecture 3 gives

$$\boxed{|N_m - (q^m + 1)| \leq 2gq^{m/2}}. \quad (24)$$

The important point is that this holds over *every* finite extension, with the same genus g .

To recall the calculation, work on $\overline{X} \times \overline{X}$, where $\overline{X} = X_{\overline{\mathbf{F}}_q}$. Let H and V be horizontal and vertical fibers, Δ the diagonal, and Γ_m the graph of the m -th iterate of q -power Frobenius. The graph has degree q^m over the second factor. Its intersections with the diagonal are exactly the points in $X(\mathbf{F}_{q^m})$, each with multiplicity one.

The corrected divisor classes

$$D_0 = \Delta - H - V, \quad D_m = \Gamma_m - H - q^m V$$

are orthogonal to both fibers. The intersection computation is

$$D_0^2 = -2g, \quad D_m^2 = -2gq^m, \quad D_0 \cdot D_m = N_m - (q^m + 1).$$

Hodge index makes $\langle D, E \rangle = -D \cdot E$ a positive-definite pairing on this subspace of numerical classes. Cauchy–Schwarz therefore gives

$$(N_m - (q^m + 1))^2 \leq (-D_0^2)(-D_m^2) = 4g^2 q^m,$$

which is (24). The genus-zero case is included: the zero norms force the mixed pairing to vanish.

This is a recall of the geometric argument, with the characteristic- p Hodge index theorem as its input. It does not use Weil's RH. Combining it with (23) gives

$$\left| \sum_{j=1}^{2g} \alpha_j^m \right| \leq 2g(\sqrt{q})^m \quad (m \geq 1). \quad (25)$$

We must now pass from a bound on every power sum to a bound on each individual reciprocal root.

6.2 A power-sum lemma: cancellation cannot persist

A small value of $\sum_j \alpha_j$ alone does not control the sizes of the α_j : two large terms might cancel. The next lemma explains why this cannot happen simultaneously for every exponent.

Lemma 6.1 (Power-sum bound). *Let $z_1, \dots, z_r \in \mathbf{C}$, and suppose that there are constants $C > 0$ and $R > 0$ such that*

$$\left| \sum_{j=1}^r z_j^m \right| \leq CR^m \quad \text{for every } m \geq 1.$$

Then $|z_j| \leq R$ for every j .

Proof. Suppose instead that

$$\rho = \max_j |z_j| > R,$$

and relabel so that $|z_1| = \rho$. In particular, $z_1 \neq 0$. Consider the arithmetic averages

$$T_M = \frac{1}{M} \sum_{m=1}^M \frac{\sum_{j=1}^r z_j^m}{z_1^m}.$$

The assumed bound gives

$$|T_M| \leq \frac{C}{M} \sum_{m=1}^M \left(\frac{R}{\rho} \right)^m \leq \frac{C}{M} \frac{R}{\rho - R} \rightarrow 0.$$

On the other hand, interchange the two finite sums:

$$T_M = \sum_{j=1}^r \frac{1}{M} \sum_{m=1}^M \left(\frac{z_j}{z_1} \right)^m.$$

For any $w \in \mathbf{C}$ with $|w| \leq 1$,

$$\frac{1}{M} \sum_{m=1}^M w^m \rightarrow \begin{cases} 1, & w = 1, \\ 0, & w \neq 1. \end{cases} \quad (26)$$

Indeed, the first case is immediate. In the second,

$$\sum_{m=1}^M w^m = \frac{w(1 - w^M)}{1 - w},$$

whose numerator stays bounded, so division by M makes the expression tend to zero.

Each z_j/z_1 has modulus at most one. Applying (26) to these finitely many ratios shows that

$$T_M \rightarrow \#\{j : z_j = z_1\}.$$

This is a positive integer, since at least $j = 1$ is counted. It contradicts the limit zero obtained above. Thus no $|z_j|$ can exceed R . $\square$

The averaging does not require the arguments of the z_j to be rational multiples of π . It also handles repeated roots: repetitions of z_1 add a positive multiplicity rather than canceling.

6.3 From root bounds to the critical line

Theorem 6.2 (Weil's RH from square-root point bounds). *Suppose that, for some constant C independent of m ,*

$$|N_m - (q^m + 1)| \leq Cq^{m/2} \quad (m \geq 1).$$

Then every reciprocal root of $L_F(t)$ satisfies

$$|\alpha_j| = \sqrt{q}. \quad (27)$$

Equivalently, every zero of the meromorphic function $\zeta_F(s) = Z_F(q^{-s})$ lies on $\operatorname{Re}(s) = 1/2$.

Proof. When $g = 0$, $L_F = 1$ and there are no zeros, so the assertion is vacuous. Assume $g \geq 1$.

First, obtain the upper bound. By (23), the hypothesis is a bound on all power sums of the α_j . The preceding lemma with $R = \sqrt{q}$ gives

$$|\alpha_j| \leq \sqrt{q} \quad (1 \leq j \leq 2g).$$

Second, use the functional equation for the reverse bound. If α is a reciprocal root, then $L_F(1/\alpha) = 0$. Substitute $t = \alpha/q$ into (19):

$$L_F\left(\frac{\alpha}{q}\right) = q^g \left(\frac{\alpha}{q}\right)^{2g} L_F\left(\frac{1}{\alpha}\right) = 0.$$

Thus α/q is an actual root, so q/α is another reciprocal root. Applying the upper bound to it gives

$$\left|\frac{q}{\alpha}\right| \leq \sqrt{q}, \quad \text{hence} \quad |\alpha| \geq \sqrt{q}.$$

This proves (27).

Finally, change variables. The actual roots of $L_F(t)$ have modulus $q^{-1/2}$. Since the denominator poles are not canceled, the zeros of $\zeta_F(s)$ are exactly the solutions of

$$q^{-s} = \alpha_j^{-1}$$

for some j . Taking absolute values gives

$$q^{-\operatorname{Re}(s)} = |\alpha_j|^{-1} = q^{-1/2}.$$

Since $q > 1$, this forces $\operatorname{Re}(s) = 1/2$. □

Applying the theorem with $C = 2g$ from Hodge index proves the root-location statement in [M, Theorem 7.20]. In the other direction, (27) immediately gives the sharp Hasse–Weil coefficient:

$$|N_m - (q^m + 1)| = \left| \sum_{j=1}^{2g} \alpha_j^m \right| \leq \sum_{j=1}^{2g} |\alpha_j|^m = 2gq^{m/2}.$$

Thus, once rationality and the functional equation are known, RH is equivalent to uniform square-root point-count bounds over all finite extensions.

Remark 6.3 (What “all zeros” means here). Write $\alpha_j = \sqrt{q} e^{i\theta_j}$. The zeros in the s -plane are

$$s = \frac{1}{2} + \frac{i(\theta_j + 2\pi k)}{\log q}, \quad k \in \mathbf{Z}.$$

The substitution $t = q^{-s}$ has imaginary period $2\pi i / \log q$. The denominator gives simple poles at

$$s = \frac{2\pi i k}{\log q} \quad \text{and} \quad s = 1 + \frac{2\pi i k}{\log q}, \quad k \in \mathbf{Z},$$

not zeros. For this projective-curve zeta function there is no additional family of trivial zeros analogous to the classical negative-even-integer zeros.

Remark 6.4 (Why the base-field estimate alone is insufficient). The case $m = 1$ bounds only $\sum_j \alpha_j$. For example, $M + (-M) = 0$ places no upper bound on $|M|$. The power-sum lemma uses the estimate for every m ; Hodge index supplies exactly that input through all iterates of Frobenius. One must not substitute a single Hasse–Weil estimate into that lemma.

6.4 The cubic from Lecture 2

Consider again the smooth projective cubic over $\mathbf{F}_3$ with affine equation

$$y^2 = x^3 + 2x + 2.$$

It has genus one and just one $\mathbf{F}_3$ -rational point, the point at infinity. The constant, linear, and leading coefficients of its L -polynomial therefore give

$$L_F(t) = 1 - 3t + 3t^2.$$

Its reciprocal roots are

$$\alpha_1 = \frac{3 + i\sqrt{3}}{2}, \quad \alpha_2 = \frac{3 - i\sqrt{3}}{2}.$$

Both have modulus $\sqrt{3}$, as required. Let $S_m = \alpha_1^m + \alpha_2^m$. Since each reciprocal root satisfies $\alpha^2 - 3\alpha + 3 = 0$,

$$S_0 = 2, \quad S_1 = 3, \quad S_m = 3S_{m-1} - 3S_{m-2} \quad (m \geq 2).$$

It follows that $S_2 = 3$, $S_3 = 0$, and $S_4 = -9$. The point-count formula gives

$$N_1 = 1, \quad N_2 = 7, \quad N_3 = 28, \quad N_4 = 91.$$

In particular, the value $N_2 = 7$ agrees with the direct computation over $\mathbf{F}_9$ in Lecture 2.

The three counts introduced at the beginning really are different. In this example,

$$B_2 = \frac{N_2 - N_1}{2} = 3, \quad A_2 = \binom{N_1 + 1}{2} + B_2 = 4,$$

whereas $N_2 = 7$. The four effective divisors of degree two are the doubled point at infinity and the three degree-two closed points.

7 The Drinfeld–Vlăduț bound

The Hasse–Weil bound is a statement about each curve. We now impose the simultaneous restrictions on its counts over all extensions to obtain a stronger bound for families of growing genus. This supplies a full averaging proof of the asymptotic theorem stated in Lecture 3 and in [W, Chapter 7]; it is additional to Morandi’s Chapter 7.

7.1 The fixed-field asymptotic problem

Recall the notation

$$N_q(g) = \max\{\#X(\mathbf{F}_q) : X/\mathbf{F}_q \text{ smooth projective geometrically integral of genus } g\},$$

and the Ihara constant

$$A(q) = \limsup_{g \rightarrow \infty} \frac{N_q(g)}{g}. \quad (28)$$

Here q is fixed while the genus grows. Dividing the usual Hasse–Weil upper bound by g gives only

$$A(q) \leq 2\sqrt{q}.$$

Our goal is the sharper inequality

$$A(q) \leq \sqrt{q} - 1.$$

By RH, write

$$\alpha_j = \sqrt{q} e^{i\theta_j}, \quad \theta_j \in \mathbf{R}.$$

The point-count formula becomes

$$\sum_{j=1}^{2g} e^{im\theta_j} = q^{-m/2}(q^m + 1 - N_m). \quad (29)$$

In particular, this sum is real, and hence

$$\sum_{j=1}^{2g} \cos(m\theta_j) = q^{m/2} + q^{-m/2} - q^{-m/2} N_m. \quad (30)$$

We also know $N_m \geq N_1$ for every m . These inequalities link the different moments: their values cannot be chosen independently.

The strategy is to combine finitely many identities (30) with positive weights. We need weights for which the corresponding trigonometric expression is nonnegative for every angle.

7.2 The nonnegative averaging kernel

Lemma 7.1 (Fejér kernel). *For an integer $M \geq 1$, define*

$$K_M(\theta) = \frac{1}{M+1} \left| \sum_{r=0}^M e^{ir\theta} \right|^2. \quad (31)$$

Then $K_M(\theta) \geq 0$ for every real θ , and

$$K_M(\theta) = 1 + 2 \sum_{m=1}^M \left(1 - \frac{m}{M+1}\right) \cos(m\theta). \quad (32)$$

Proof. Nonnegativity is immediate from the squared absolute value. Expand that square:

$$\left| \sum_{r=0}^M e^{ir\theta} \right|^2 = \sum_{r=0}^M \sum_{s=0}^M e^{i(r-s)\theta}.$$

There are $M+1$ terms with $r-s=0$. For each $1 \leq m \leq M$, there are $M+1-m$ terms with $r-s=m$, and the same number with $r-s=-m$. Pairing these terms and using

$$e^{im\theta} + e^{-im\theta} = 2\cos(m\theta)$$

gives

$$(M+1) + 2 \sum_{m=1}^M (M+1-m) \cos(m\theta).$$

Divide by $M+1$. □

We will use the positive weights

$$c_{M,m} = 1 - \frac{m}{M+1} \quad (1 \leq m \leq M).$$

For each fixed m , these weights tend to 1 as $M \rightarrow \infty$. Their purpose is not merely to average the point counts: they ensure the nonnegativity in (31), which an arbitrary unweighted finite cosine sum would not have.

7.3 A finite-genus inequality

Proposition 7.2 (Weighted point-count inequality). *For every $M \geq 1$ and every curve $X/\mathbf{F}_q$ of genus g ,*

$$\sum_{m=1}^M c_{M,m} q^{-m/2} N_m \leq g + \sum_{m=1}^M c_{M,m} (q^{m/2} + q^{-m/2}). \quad (33)$$

Define the positive quantities

$$D_M(q) = \sum_{m=1}^M c_{M,m} q^{-m/2}, \quad (34)$$

$$E_M(q) = \sum_{m=1}^M c_{M,m} (q^{m/2} + q^{-m/2}). \quad (35)$$

Then

$$\boxed{N_1 \leq \frac{g + E_M(q)}{D_M(q)}}. \quad (36)$$

Proof. Evaluate the nonnegative kernel at all $2g$ angles and sum. By (32) and (30),

$$\begin{aligned} 0 &\leq \sum_{j=1}^{2g} K_M(\theta_j) \\ &= 2g + 2 \sum_{m=1}^M c_{M,m} \sum_{j=1}^{2g} \cos(m\theta_j) \\ &= 2g + 2 \sum_{m=1}^M c_{M,m} (q^{m/2} + q^{-m/2} - q^{-m/2} N_m). \end{aligned}$$

Move the point-count terms to the other side and divide by 2. This proves (33).

Every $c_{M,m}q^{-m/2}$ is positive and $N_m \geq N_1$. Therefore

$$N_1 D_M(q) \leq \sum_{m=1}^M c_{M,m} q^{-m/2} N_m \leq g + E_M(q).$$

Since $D_M(q) > 0$, division proves (36). $\square$

Example 7.3 (The first kernel recovers Hasse–Weil). For $M = 1$, the only weight is $c_{1,1} = 1/2$, so

$$D_1(q) = \frac{1}{2\sqrt{q}}, \quad E_1(q) = \frac{1}{2} \left(\sqrt{q} + \frac{1}{\sqrt{q}} \right).$$

Formula (36) becomes

$$N_1 \leq 2g\sqrt{q} + q + 1,$$

exactly the Hasse–Weil upper bound. Larger M uses more extension-field information. The asymptotic improvement will come from letting M grow *after* taking the large-genus limit.

7.4 Taking the limits in the correct order

Theorem 7.4 (Drinfeld–Vlăduț). *For every prime power q ,*

$$A(q) = \limsup_{g \rightarrow \infty} \frac{N_q(g)}{g} \leq \sqrt{q} - 1. \quad (37)$$

Proof. First fix M and let the genus grow. The constants $D_M(q)$ and $E_M(q)$ depend only on q and M , not on the curve. Thus (36) holds uniformly over all curves of genus g and gives, for $g \geq 1$,

$$\frac{N_q(g)}{g} \leq \frac{1}{D_M(q)} + \frac{E_M(q)}{gD_M(q)}.$$

The second term tends to zero as $g \rightarrow \infty$. Therefore, for every fixed M ,

$$A(q) \leq \frac{1}{D_M(q)}. \quad (38)$$

Now let M grow. Put $\eta = q^{-1/2}$, so $0 < \eta < 1$. We have

$$D_M(q) = \sum_{m=1}^M \left(1 - \frac{m}{M+1} \right) \eta^m.$$

To identify its limit explicitly, compare with the geometric series:

$$\begin{aligned} 0 \leq \sum_{m=1}^{\infty} \eta^m - D_M(q) &= \sum_{m>M} \eta^m + \frac{1}{M+1} \sum_{m=1}^M m\eta^m \\ &\leq \frac{\eta^{M+1}}{1-\eta} + \frac{\eta}{(M+1)(1-\eta)^2} \rightarrow 0. \end{aligned}$$

Here $\sum_{m \geq 1} m\eta^m = \eta/(1-\eta)^2$, obtained by differentiating the convergent geometric series. Consequently,

$$D_M(q) \rightarrow \frac{\eta}{1-\eta} = \frac{1}{\sqrt{q}-1}.$$

Taking $M \rightarrow \infty$ in (38) proves (37). $\square$

Remark 7.5 (The order of limits matters). We do not let $M \rightarrow \infty$ in (36) while keeping g fixed: the terms $q^{m/2}$ make $E_M(q)$ unbounded. Instead, for each fixed M , first divide by g and let $g \rightarrow \infty$, removing $E_M(q)/g$. Only then increase M .

Equivalently, given $\varepsilon > 0$, first choose M so that

$$\frac{1}{D_M(q)} \leq \sqrt{q} - 1 + \frac{\varepsilon}{2}.$$

Then choose g large enough that $E_M(q)/(gD_M(q)) \leq \varepsilon/2$. For all such genera, uniformly in the curve,

$$\frac{N_q(g)}{g} \leq \sqrt{q} - 1 + \varepsilon.$$

This is the precise large-genus assertion.

No step of the upper-bound proof assumes that q is a square. The square-field hypothesis enters the constructions attaining the bound, not this inequality.

For example, over $\mathbf{F}_{49}$, Hasse–Weil gives $N_1 \leq 50 + 14g$, whereas Drinfeld–Vlăduț says that for every $\varepsilon > 0$,

$$N_{49}(g) \leq (6 + \varepsilon)g$$

for all sufficiently large g . It does not assert $N_1 \leq 6g$ for every individual curve. A sharp bound at one fixed genus and the sharp asymptotic ratio are different questions.

The proof has used three pieces of information together: the exact formula for N_m , the RH constraint $|\alpha_j| = \sqrt{q}$, and the persistence $N_m \geq N_1$ of base-field rational points. The positive kernel combines them without discarding the relations among the different extension degrees.

8 Next time

TVZ–Ihara: attaining the upper bound. For a square prime power q , we will turn to the construction of families $X_i/\mathbf{F}_q$ with

$$g_i \rightarrow \infty, \quad \frac{\#X_i(\mathbf{F}_q)}{g_i} \rightarrow \sqrt{q} - 1.$$

This is the existence theorem stated in Lecture 3; together with today’s upper bound it yields $A(q) = \sqrt{q} - 1$. The AG-code argument from that lecture then gives the TVZ lower bound

$$\alpha_q(\delta) \geq 1 - \delta - \frac{1}{\sqrt{q} - 1}.$$

Today’s theorem supplies the obstruction; the next construction supplies families reaching it.

A symplectic interpretation of $L(D)$. We will also return to the function spaces themselves. The identity

$$L(D) = H^0(X, \mathcal{O}_X(D)) = \text{Hom}_X(\mathcal{O}_X, \mathcal{O}_X(D))$$

suggests a new question: in a homological-mirror-symmetry setting, how is this morphism space interpreted on the symplectic side? The geometric setting and the corresponding Floer-theoretic interpretation will be developed separately; no such interpretation is used in today’s finite-field proofs.

Lecture summary

The zeta function has three compatible expressions:

$$Z_F(t) = \sum_{E \geq 0} t^{\deg E} = \prod_{P \text{ closed}} \frac{1}{1 - t^{\deg P}} = \exp \left(\sum_{m \geq 1} \frac{\#X(\mathbf{F}_{q^m})}{m} t^m \right).$$

Riemann–Roch and divisor-class counting give

$$Z_F(t) = \frac{L_F(t)}{(1-t)(1-qt)}, \quad L_F(t) = \prod_{j=1}^{2g} (1 - \alpha_j t), \quad L_F(t) = q^g t^{2g} L_F\left(\frac{1}{qt}\right).$$

The resulting point-count identity is

$$N_m = q^m + 1 - \sum_{j=1}^{2g} \alpha_j^m.$$

The two main deductions are:

- The Hodge-index proof from Lecture 3 bounds these power sums for every m . The averaging lemma gives $|\alpha_j| \leq \sqrt{q}$, and the functional equation gives the reverse inequality. Thus the actual roots in the t -plane have modulus $q^{-1/2}$, and all zeros of $\zeta_F(s)$ lie on $\operatorname{Re}(s) = 1/2$.
- Nonnegativity of the Fejér kernel, combined with $N_m \geq N_1$, gives the finite inequality $N_1 D_M(q) \leq g + E_M(q)$. First letting $g \rightarrow \infty$ and then $M \rightarrow \infty$ yields $A(q) \leq \sqrt{q} - 1$.

The logical dependencies are therefore

$$\begin{aligned} \text{Riemann–Roch} &\implies \text{rationality and functional equation,} \\ \text{Hodge index} &\implies \text{Hasse–Weil over all finite extensions} \implies \text{Weil’s RH,} \\ \text{RH, extension-field counts, and positive averaging} &\implies \text{Drinfeld–Vlăduț.} \end{aligned}$$

References

- [M] P. J. Morandi, *Error Correcting Codes and Algebraic Curves*, lecture notes for Mathematics 601, Fall 2001. For this lecture: Chapter 7, especially §§7.1–7.3, Lemmas 7.2 and 7.5, Proposition 7.12 and Corollary 7.13, Proposition 7.16, and Theorem 7.18. Theorem 7.20 is the RH statement, and Corollary 7.21 is the Hasse–Weil bound. The Euler-factor sign and zero-divisor normalization are specified in the text above.
- [W] J. L. Walker, *Codes and Curves*, Student Mathematical Library, vol. 7, American Mathematical Society. For this lecture: Chapter 7, on point counts, the Ihara constant, the Drinfeld–Vlăduț bound, and the TVZ–Ihara existence theorem and coding bound.